Benjamin Ellis-Bloor

Email: benjaminellis-bloor2026@u.northwestern.edu
Website: https://ellis-bloor.github.io

Education

Northwestern University — Evanston, IL
PhD in Mathematics — *2021 – 2027 (Expected)*
- Advisor: Ezra Getzler

Australian National University — Canberra, ACT
PhB (Honours) (First Class) in Mathematics — *2017 – 2020*
- ANU National Merit Scholarship

Papers

Cobordism-valued intersection theory on $\overline{\mathcal{M}}_{0,n}$, *submitted,* arXiv:2603.03829 — 2026
Localization, cobordism, and twisted χ_y-genera of $\overline{\mathcal{M}}_{0,n}$, *preprint,* arXiv:2610.37628 — 2026

Talks

Northwestern Algebraic Geometry Seminar — November 2026
ETH Zürich Algebraic Geometry and Moduli Seminar — September 2026
Northwestern Informal Algebraic Geometry Oberseminar — October 2025
Northwestern Informal Algebraic Geometry Oberseminar — April 2025

Academic Awards

Chancellor's Letter of Commendation — 2019
Dean's Science Education Commendation — 2019
Hanna Neumann Prize for Second Year Mathematics — 2018
Chancellor's Letter of Commendation — 2018
Dean's Science Education Commendation — 2018
Chancellor's Letter of Commendation — 2017

Teaching Experience

Teaching Assistant — 2022-Present
Northwestern University
- Math 218-1: Single-Variable Calculus with Precalculus (Fall 2025)
- Math 300-0: Foundations of Higher Mathematics (Spring 2025)
- Math 320-1: Real Analysis (Spring 2025, Fall 2024)
- Math 334-0: Linear Algebra: Second Course (Spring 2024)
- Math 290-1,2,3: MENU: Linear Algebra and Multivariable Calculus (Fall 2024, Spring 2024, Winter 2024, Fall 2023)
- Math 240-0: Linear Algebra (Winter 2024, Fall 2023)
- Math 100-0: Quantitative Reasoning (Spring 2023)
- Math 230-2: Multivariable Integral Calculus (Spring 2023)
- Math 230-1: Multivariable Differential Calculus (Winter 2023)
- Math 220-1: Single-Variable Differential Calculus (Fall 2022)

Tutor — 2021
University of Sydney
- MATH1921: Calculus of One Variable (Advanced) (Semester 1 2021)
- MATH1023: Multivariable Calculus and Modelling (Semester 1 2021)

Demonstrator — 2019-2021
Australian National University
- MATH1115: Advanced Mathematics and Applications 1 (Semester 1 2021)
- MATH1113: Mathematical Foundations for Actuarial Studies (Semester 2 2020)
- MATH1005: Discrete Mathematical Models (Semester 1 2020, Semester 1 2019)